Effects of Nitrogen Availability on Early Growth of Arabidopsis thaliana

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Abstract
Nitrogen is an essential macronutrient that strongly influences plant growth and development. In this study, we investigated the effect of three nitrogen concentrations on early growth parameters of *Arabidopsis thaliana*. Seeds were grown under controlled conditions and supplied with low, medium, or high nitrogen nutrient solutions. After 21 days, biomass accumulation, leaf number, and chlorophyll content were measured. Plants grown under medium nitrogen conditions showed optimal growth, while both low and high nitrogen levels resulted in reduced performance. These results demonstrate the importance of balanced nitrogen availability during early plant development.

Introduction
Nitrogen plays a central role in plant metabolism as a component of amino acids, nucleic acids, and chlorophyll. Both nitrogen deficiency and excess can negatively affect plant growth, leading to reduced productivity and physiological stress. *Arabidopsis thaliana* is a widely used model organism in plant biology due to its short life cycle and well-characterized genome. The aim of this study was to assess how different nitrogen concentrations influence early growth traits of *A. thaliana* under controlled environmental conditions.

Materials and Methods
Biological Material. Seeds of *Arabidopsis thaliana* ecotype Columbia-0 (Col-0) were used in all experiments.

Experimental Design. The experiment followed a completely randomized design with three nitrogen treatments: low nitrogen (0.5 mM nitrate), medium nitrogen (5 mM nitrate), and high nitrogen (15 mM nitrate). Each treatment consisted of 10 biological replicates, resulting in a total of 30 plants.

Growth Conditions. Seeds were surface-sterilized and stratified at 4 °C for 2 days. Plants were grown in 7 cm pots filled with sterile vermiculite in a controlled growth chamber (16 h light / 8 h dark, 150 µmol m■² s■¹, 22/20 °C day/night, 60% relative humidity). Plants were irrigated every two days with modified Hoagland solution adjusted to the appropriate nitrogen concentration.

Measurements. After 21 days, rosette leaf number, fresh weight, and chlorophyll content (SPAD meter) were recorded. Fresh weight was determined immediately after harvesting. Chlorophyll measurements were taken from the youngest fully expanded leaf.

Data Analysis. Mean values and standard deviations were calculated for each treatment. Differences among treatments were evaluated using one-way analysis of variance (ANOVA).

Results
Plants grown under medium nitrogen conditions displayed the highest fresh weight and leaf number. Low nitrogen plants exhibited pale leaves and reduced biomass, while high nitrogen plants showed slightly reduced growth and occasional leaf curling. Chlorophyll content followed a similar pattern, with the highest SPAD values observed in the medium nitrogen group.

Discussion

The results indicate that moderate nitrogen availability supports optimal early growth of *A. thaliana*. Nitrogen deficiency limited chlorophyll synthesis and biomass accumulation, while excessive nitrogen may have caused metabolic imbalance. These findings are consistent with previous reports highlighting the importance of nitrogen homeostasis in plants.

Conclusion

This study demonstrates that nitrogen concentration significantly affects early plant growth. The medium nitrogen treatment promoted the most favorable growth characteristics, emphasizing the need for balanced nutrient supply during early developmental stages.